Synthetic algebraic geometry in Unit: what has been formalised

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Abstract

We report on a first formalisation of parts of *A foundation for synthetic algebraic geometry* (Cherubini, Coquand, Hutzler [1]) in the prototype Unit, the version where recursion is only through eliminators. The foundation is a univalent universe with Rijke's axiom of replacement, as formulated in [4]; propositional truncation and set quotients are derived from it. Finitely presented algebras are given by presentations and a universal property, and every presentation has a presented algebra. SQC is one uniform hypothesis on all finitely presented algebras. The results checked are Lemma 1.3.8, Propositions 2.2.2–2.2.3, Lemma 2.2.4 (from Loc and SQC only), Sections 3 and 4 (including two proofs of Theorem 3.3.3), Section 5 under the full SAG axioms (gluing has its universal property for small set targets), Example 6.1.9 (the fibres computed in the example, and the coordinate algebra of a fibre), Lemma 6.2.1, Section 7.1 except Remark 7.1.3 (the localised module, 7.1.2 and 7.1.8 as equivalences of modules, and 7.1.9–7.1.13 under the full SAG axioms), Lemma 7.3.5, Theorem 7.3.6 (from the SAG axioms and fibrewise weak quasi-coherence), Lemma 7.4.2, Theorem 7.4.7 and Theorem 7.4.8. From *Projective space in synthetic algebraic geometry* [2], Proposition 4.6 is checked: line bundles on $\mathbb{P}^1$ are $\mathsf{Line}\times\mathbb{Z}$. It uses the rank-one case of Horrocks' theorem, proved for any commutative ring. Also checked are the two definitions of $\mathbb{P}^n$ and their equivalence, Lemma 1.6, Proposition 1.7, and Section 7 of [2]: line bundles on $\mathbb{P}^n$, $n \geq 2$, are $\mathsf{Line} \times \mathbb{Z}$; and Corollary 5.6: $\mathrm{Aut}(\mathbb{P}^n) = \mathrm{PGL}_{n+1}(R)$ for $n \geq 2$. The combined development through Section 5 checks in 1.30 seconds.

1 The foundation

All developments share one segment whose parameters are the foundational axioms:

- *univalence*, at every level, in the form "`idtoeqv` is an equivalence";
- *replacement*, as two parameters which together form one axiom schema [4, Definitions 2, 4, 6]:
  - `images`: for $A : \mathsf{U}_l$, a locally U_l-small $B : \mathsf{U}_k$ and $f : A \to B$, a chosen factorisation $A \to I \to B$ with $I : \mathsf{U}_l$, a commuting triangle, and $I \to B$ an embedding;
  - `surjections`: the map $A \to I$ is surjective in the sense that precomposition is an equivalence on sections of every family of propositions over I, at every level j.

The chosen image does not depend on the level j of elimination. The schema is split into data and a property only because a type quantifying over all levels does not belong to a universe. No resizing is assumed.

From these parameters the following are derived, not assumed:

- function extensionality, from univalence;

- the propositional truncation $\|A\|$, as the image of $A \to 1$, with induction into propositions of any level and its computation rule as an identity;
- set quotients of proposition-valued equivalence relations, as the image of $a \mapsto (x \mapsto R(x,a))$ into $A \to \mathsf{Prop}$ (locally small by univalence); they are effective, and maps into sets that respect the relation descend to them;
- univalent completions of small families.

The computation rules of the derived truncation and quotients are proved identities. The evaluator has no new reduction rules.

2 How the mathematics is represented

Rings. A commutative ring is a record with a carrier, the operations, and the axioms as identities. Where needed the carrier is assumed to be a set. A unit is an element with a chosen inverse, which is a proposition when the carrier is a set.

Polynomials. A polynomial is a list of coefficients, lowest degree first. Two lists are equal as polynomials when all their coefficients are equal, so trailing zeros do not matter. In the files on $\mathbb{P}^1$ the type $R[X]$ is the quotient of lists by this relation; in the file on Lemma 2.2.4 we keep lists and the relation, which suffices. Evaluation is Horner's rule, a primitive recursion. Laurent polynomials are pairs of lists (p, n), standing for $p(X) + X^{-1}n(X^{-1})$.

Recursion. All recursive definitions and all proofs by induction are written with the eliminators of the data types, as in system T but with dependent types. Several proofs use recursion at higher type. For instance, the induction on the degree in Lemma 2.2.4 returns, at each step, a function over all polynomials of the smaller length.

Finitely presented algebras. An R-algebra is a commutative ring whose carrier is a set, with a ring homomorphism from R. A presentation is a number n and a list of expressions in the variables $\mathsf{Fin}\, n$; expressions are an ordinary data type (variables, constants of R, sums, products), and Fin is defined by recursion ($\mathsf{Fin}\, 0 = \emptyset$, $\mathsf{Fin}(n+1) = \mathsf{Fin}\, n + 1$), so no indexed families are needed. The solutions of P in an algebra B are the assignments $V \to B$ on which every equation evaluates to 0. An algebra A is *presented* by P when it has a solution a of P such that for every algebra B the map $\mathrm{Hom}(A,B) \to \mathrm{Sol}_P(B)$, $\varphi \mapsto \varphi \circ a$, is an equivalence; A is *finitely presented* when merely there is a presentation of it, as in [1, Definition 1.3.2]. Then $\mathrm{Spec}A = \mathrm{Hom}(A,R)$ is equivalent to $\mathrm{Sol}_P(R)$, the solutions of the equations in R. Every presentation has a presented algebra: expressions modulo the relation of merely having a finite derivation from the ring laws and the equations. A derivation is an ordinary tree with a validity predicate defined by recursion. Two algebras with the same presentation are isomorphic. The polynomial ring $A[X]$, built as a quotient of lists, has the universal property $\mathrm{Hom}(A[X],B) \simeq \mathrm{Hom}(A,B) \times B$, and is presented by any presentation of A with one more variable.

The axioms of [1]. Loc, SQC and Z-choice are not part of the foundation. When a proof needs one of them, it is a hypothesis of the segment containing the proof. SQC is stated once: for every finitely presented algebra A, evaluation $A \to (\mathrm{Spec}A \to R)$ is an equivalence. Some earlier files use instances of SQC for particular algebras instead; this is said for each result. Z-choice is used only

in Theorem 7.3.6, where it is a hypothesis for one affine X, together with weak quasi-coherence of the module.

3 The results

Proposition 2.2.2: the weak Nullstellensatz

For a finitely presented A, $\mathrm{Spec}A = \emptyset$ if and only if $1 = 0$ in A. By SQC, an empty spectrum gives $A \simeq (\emptyset \to R)$, a one-element ring; the converse uses $1 \neq 0$ in R.

For $A = R[X]/(f)$, presented by one variable and the equation f, the algebraic content is a certificate: if $1 = 0$ in A then merely $fg = 1$ for some list g, with equality of coefficients. It is obtained by reading a finite derivation of $1 = 0$ as a computation of a multiplier: each rule (the ring laws, the equation $f = 0$, symmetry, transitivity, compatibility with $+$ and $\times$) transforms multipliers r with $x = y + fr$. No second quotient of $R[X]$ by an ideal is built, and SQC is not used for this step. Combined with SQC: if f has no root then merely $fg = 1$ for some g.

Lemma 2.2.4: R is algebraically closed up to double negation

For a polynomial f, it is not not the case that $f = 0$ or

$$f = \alpha\,(X - a_1)^{e_1} \cdots (X - a_n)^{e_n},$$

with α a unit, $e_i \geq 1$ and the a_i pairwise distinct.

The proof follows the paper: induction on a bound for the degree, inside the double negation monad. The top coefficient is zero (drop it) or not. In the second case it is a unit, and if the degree is at least 1 there is not not a root a. Then synthetic division by $X - a$ gives $f = (X - a)q$, and q has a smaller bound. Synthetic division is one primitive recursion returning both the quotient and the remainder, and gives the factor theorem with the bound on the degree. A last induction groups equal roots, deciding $a_i = a_j$ under the double negation. This uses that $X - a$ and $X - b$ commute, a coefficient computation.

Hypotheses: Loc and SQC. The proof was first written with two hypotheses, $1 \neq 0$ and the weak Nullstellensatz for $R[X]/(f)$ in the form: if f has no root then it is not not the case that $fg = 1$ for some list g. Double negation suffices because the proof only uses it inside double negations, and the mere inverse given by Proposition 2.2.2 implies it. With Proposition 2.2.2 the lemma follows from Loc and SQC alone. From the Nullstellensatz we prove Proposition 2.2.3(a) up to double negation (a constant $x \neq 0$ has no root, hence is not not invertible). We also prove that a polynomial of degree ≥ 1 with a unit top coefficient is not invertible: the top coefficient of fg is the product of the top coefficients. Replacement is not needed here. Nine computations on $\mathbb{Z}/2$ check that the definitions compute, for instance $1 + X^2 = (X + 1)^2$.

Section 3.1: affine types, localisation and affine opens

An affine witness for X consists of an R-algebra A, mere finite presentability of A, and an equivalence $X \simeq \mathrm{Spec}A$; affineness is its propositional truncation (Definition 3.1.1). The pointwise ring R^X is constructed as an actual R-algebra. Under SQC, evaluation at points gives the canonical equivalence $X \simeq \mathrm{Spec}(R^X)$ (Lemma 3.1.2). For Proposition 3.1.3, both representing algebras are compared with R^X. A structure identity lemma, proved from univalence, retains the X-indexed points as well as the ring operations and constants. It therefore identifies the entire affine witnesses, including their spectrum identifications. Their type is a proposition. In particular, R^X is finitely presented when X is affine.

The algebraic constructions. Adjoining a variable y with equation $fy - 1$ constructs $A[f^{-1}]$, with finite presentability and its canonical universal property for every target R-algebra B:

$$\mathrm{Hom}(A[f^{-1}], B) \simeq \sum_{h:\mathrm{Hom}(A,B)} \mathrm{inv}(h(f)).$$

At $B = R$ this gives $D(f) = \mathrm{Spec}A[f^{-1}]$ (Proposition 3.1.6). Appending finitely many equations constructs $A/(g_1, \ldots, g_n)$ with the analogous universal property and spectrum the simultaneous zero locus. Elements of a presented algebra are merely expressions in its generators. Those expressions, and finite presentations themselves, are eliminated only into propositions: no presentation is chosen from mere finite presentability.

Opens and certificates. Uniform SQC and $1 \neq 0$, applied to $R/(a)$, imply $a \neq 0 \Rightarrow \mathrm{inv}(a)$; the converse is elementary. This gives Example 3.1.4 and the equivalence between the two descriptions of affine-open subtypes in Definition 3.1.5. Every element of $A[f^{-1}]$ is merely a fraction h/f^k. The open it defines is the restriction of $D(hf)$, which is contained in $D(f)$. Finite choice of fractions proves Lemma 3.1.7, stated also as affine-openness of the direct image under $\mathrm{Spec}A[f^{-1}] \to \mathrm{Spec}A$. No geometric axiom is needed for this transitivity proof.

For Lemma 3.1.8, collapse of $A[f^{-1}]$ gives a polynomial certificate in $A[Y]/(fY-1)$. Comparing coefficients extracts an exponent with $f^k = 0$. With SQC and $1 \neq 0$, this proves $D(f) = \emptyset$ if and only if f is nilpotent. For Lemma 3.1.9, quotient by the g_i, apply Lemma 3.1.8 to the image of f, and interpret the quotient's finite derivation trees as ideal certificates. The result is the equivalence

$$D(f) \subseteq \bigcup_i D(g_i) \quad \Longleftrightarrow \quad \left\| \sum_{k:\mathbb{N}} \sum_{c:\mathsf{Fin}(n)\to A} f^k = \sum_i g_i c_i \right\|.$$

The reverse direction uses Loc to find an invertible summand. Thus the combined formal statement has explicit Loc and uniform SQC hypotheses. The final corollary is also proved: the $D(g_i)$ cover $\mathrm{Spec}A$ exactly when the g_i generate the unit ideal.

Sections 3.2 and 3.3.1–3.3.2: tensor products, pullbacks, eventually vanishing sequences

Lemma 3.2.1. The tensor product $A \otimes B$ is the coproduct of R-algebras: maps $A \to T$, $B \to T$ such that $h \mapsto (h \circ \iota_A, h \circ \iota_B)$ is an equivalence $\mathrm{Hom}(T, D) \simeq \mathrm{Hom}(A, D) \times \mathrm{Hom}(B, D)$ for every D. With $D = R$, $\mathrm{Spec}T = \mathrm{Spec}A \times \mathrm{Spec}B$. For A, B presented by p and q, the algebra presented by the concatenation of p and q (variables of both, equations of both, renamed) is a coproduct; so finitely presented algebras have a finitely presented tensor product. No SQC is used.

Lemma 3.2.2. For $f : C \to A$ and $g : C \to B$, add to the concatenated presentation one equation $e_v - e'_v$ for each generator c_v of C, where e_v, e'_v are expressions in the generators for $f(c_v)$ and $g(c_v)$. The algebra presented is a pushout, and its spectrum is the pullback $\mathrm{Spec}A \times_{\mathrm{Spec}C} \mathrm{Spec}B$. The expressions exist merely: every element of a presented algebra is merely the value of an expression, and finitely many can be chosen at once. By SQC every map $\mathrm{Spec}A \to \mathrm{Spec}C$ comes from a unique algebra map $C \to A$, so the pullback of any two maps of affine schemes is affine.

Lemma 3.3.1. For a tensor product T of A and B, with T and B finitely presented, $c \mapsto (\varphi \mapsto (\varphi \otimes B)(c))$ is an equivalence $T \simeq (\mathrm{Spec}A \to B)$. The proof is the paper's chain: SQC for T, Lemma 3.2.1, currying, SQC for B, using $\psi \circ (\varphi \otimes B) = \varphi \otimes \psi$.

Lemma 3.3.2. $A[X]$, the polynomial ring of the polynomial library, is a tensor product of A and $R[X]$. An eventually vanishing sequence is the sequence of coefficients of a unique element of $R[X]$; uniqueness lets us build it from a mere bound, without choice. Lemma 3.3.1 with $B = R[X]$ turns the family into one polynomial p over A, and $x \otimes R[X]$ applies x to its coefficients. The length of a list representing p is the common bound.

Theorem 3.3.3: functions $\mathrm{Spec}A \to \mathbb{N}$ are bounded

For a finitely presented A and $f : \mathrm{Spec}A \to \mathbb{N}$, merely there is N with $f(x) \leq N$ for all x. The paper's proof goes through tensor products (Lemma 3.3.1) and eventually vanishing sequences. A shorter route uses only $A[T]$:

- the function $H(x,t) = t^{f(x)}$ on $\mathrm{Spec}A \times R$ is a function on $\mathrm{Spec}(A[T])$, since a homomorphism $A[T] \to R$ is a point of $\mathrm{Spec}A$ and the image of T;
- by SQC for $A[T]$, H is merely the evaluation of one polynomial p over A;
- at a point x, the polynomial obtained by applying x to the coefficients of p has the same values as $T^{f(x)}$, hence the same coefficients (SQC for $R[T]$);
- if $f(x)$ is at least the length of p, its coefficient at $f(x)$ is 0, while that of $T^{f(x)}$ is 1; this contradicts $1 \neq 0$.

The bound is the length of p. It is a mere bound because p is merely a list. Order on $\mathbb{N}$ is decidable, so no excluded middle is used. There are two versions: one with $1 \neq 0$ and the two halves of SQC used (surjectivity for $A[T]$, injectivity for $R[T]$) as hypotheses, and one from $1 \neq 0$ and uniform SQC, where $A[T]$ and $R[T]$ are shown finitely presented.

The paper's proof is also formalised: it applies Lemma 3.3.2 to the sequences equal to 1 below $f(x)$ and 0 from $f(x)$ on, and uses $1 \neq 0$. It has the same hypotheses as the uniform version of the shorter proof: uniform SQC, $1 \neq 0$ and finite presentation of A.

Proposition 3.3.4: upwards closed families have a common element

The axioms of Section 2.1 are now stated uniformly: Loc, SQC and Z-choice, with an independent universe for the family in Z-choice. For an upwards closed proposition-valued family $P(x,n)$ on $\mathrm{Spec}A \times \mathbb{N}$, Z-choice turns pointwise mere inhabitants into finitely many local sections on $D(f_i)$, where the f_i form a unimodular family. Proposition 3.1.6 and Theorem 3.3.3 merely bound each section. Finite choice combines the bounds; their sum bounds them all. The covering property and upward closure give

$$\left(\prod_x \left\| \sum_n P(x,n) \right\|\right) \longrightarrow \left\| \sum_N \prod_x P(x,N) \right\|.$$

There is no choice over the points of $\mathrm{Spec}A$. The implementation uses the shorter proof of Theorem 3.3.3.

The combined entry point `affine/Section3.seg` checks both developments together: 2646 declarations, no failures, in 0.83 seconds. Integration lemmas turn the tensor and arbitrary-map pullback constructions into the intrinsic predicate of affineness.

Section 4.1: closed subtypes and gluing ideals

A closed proposition is merely a finite system of equations over R: an equivalence with $\bigwedge_i x_i = 0$ (Definition 4.1.1). Ideals are proposition-valued predicates with the usual closure laws; finite

generation is mere existence of a finite family generating the predicate. Membership in a generated ideal is the truncation of an explicit linear-combination certificate.

Proposition 4.1.2 (SQC). Finitely generated ideals of R are order-reversingly equivalent to closed propositions. The forward map sends I to "every member of I is zero", proved equivalent to $I = (0)$. The inverse sends a closed proposition Q to the ideal

$$\{a : R \mid Q \to a = 0\}.$$

This constructs the predicate without choosing an equation witness from a truncation. Both orders are partial orders on sets, and the equivalence reverses the orders.

Lemma 4.1.3 (SQC). Inclusion $V(f_1, \ldots, f_n) \subseteq V(g_1, \ldots, g_m)$ is equivalent to $(g_1, \ldots, g_m) \subseteq (f_1, \ldots, f_n)$ as actual ideals. Indeed, SQC on $A/(f_1, \ldots, f_n)$ detects that the images of the g_j vanish; quotient certificates then recover their ideal membership. This proof does not pass to radicals. A regression using dual numbers checks the distinction: $\epsilon^2 = 0$, but ϵ is not in the zero ideal; under SQC, $V(0)$ is not contained in $V(\epsilon)$.

Lemma 1.3.8: the algebraic gluing step. First, from the canonical localisation universal property, we prove

$$a/1 = 0 \text{ in } A_f \quad \iff \quad \left\| \sum_{k:\mathbb{N}} f^k a = 0 \right\|,$$

and its ideal-membership version

$$a/1 \in (g_j/1) \quad \iff \quad \left\| \sum_{k:\mathbb{N}} f^k a \in (g_j) \right\|.$$

The first uses a polynomial certificate in $A[Y]/(fY - 1)$ and coefficient comparison; the second combines a quotient with this localisation model. Neither needs a geometric axiom.

For compatible finitely generated ideals in the A_{f_i}, replace local generators by numerators. Clear denominators on each overlap, combine the finitely many exponents, and concatenate the numerators multiplied by these powers of f_i. The resulting ideal restricts to each prescribed ideal. Ideal extension is proved independent of generating witnesses. The final statement uses literal equalities of extended ideals. Iterated localisations are shown to satisfy the universal property of $A_{f_i f_j}$. The ring A is arbitrary, and the theorem does not require that the $D(f_i)$ cover; there is no uniqueness claim in that generality. Localisation witnesses carry finite presentability over A, which is derived from their universal property rather than assumed separately.

Lemma 4.1.4. For a pointwise closed subtype C of Spec$A$, Z-choice gives local finite equation witnesses on a finite unimodular cover. Boundedness of their lengths allows padding with zero equations; SQC turns the coordinates into elements of the local algebras. On overlaps their zero loci agree, so Lemma 4.1.3 gives agreement of their ideals. Lemma 1.3.8 glues these to a finitely generated ideal of A. The cover then identifies the global zero locus with C. The checked result retains a quotient $Q = A/(g_1, \ldots, g_n)$ with its universal property and an equivalence

$$\sum_{x:\mathrm{Spec}A} C(x) \simeq \mathrm{Spec}Q$$

over Spec$A$. A separate theorem states that a closed subtype of any affine type is affine. The hypotheses are Loc, uniform SQC and Z-choice. There is no choice over the points of Spec$A$.

The combined entry point `closed/Section4.seg` also checks the open subtype development below, with the same definition of closed propositions: 3860 declarations, no failures, in 1.06 seconds.

Section 4.2: open subtypes

A proposition is open if it is merely equivalent to one of finitely many elements of R being invertible (Definition 4.2.1); this has the same form as closed propositions, with invertibility in place of vanishing.

Proposition 2.2.3(b),(c). A vector is nonzero if and only if one of its entries is invertible: $R/(x_1, \ldots, x_n)$ has no points, so it collapses by Proposition 2.2.2, its kernel certificate gives $1 = \sum x_i c_i$, and Loc finds an invertible summand. An element is nilpotent if and only if it is not not zero, by Lemma 3.1.8 for $A = R$.

Propositions 4.2.2–4.2.3, Lemma 4.2.4. Open propositions contain false and true, are closed under binary unions (append the lists) and intersections (all products), and pull back along maps; no axiom is used. By 2.2.3(b) they are the negations of closed propositions, hence double-negation stable (Loc, SQC).

Lemmas 4.2.5–4.2.6. On an affine X, $x \neq y$ is open: a presentation embeds $\mathrm{Spec}A$ in R^n, and $x \neq y$ says that the vector of coordinate differences is nonzero. So x specialises to y if and only if $\neg\neg(x = y)$.

Theorem 4.2.7 (Loc, SQC, Z-choice). An open subtype S of $\mathrm{Spec}A$ is affine-open. As in the paper: Z-choice gives open witnesses on a cover $D(f_i)$; their lengths are bounded on $D(f_i) = \mathrm{Spec}A[f_i^{-1}]$ by Theorem 3.3.3; padding with zeros and SQC for $A[f_i^{-1}]$ give elements g_{ij} with $S \cap D(f_i) = D(g_{ij})_j$; Lemma 3.1.7 brings these down to $\mathrm{Spec}A$, and the lists are concatenated. The last step is a gluing lemma used again below: a subtype covered by finitely many $D(f_i)$ and affine-open on each $\mathrm{Spec}A[f_i^{-1}]$ is affine-open.

Lemmas 4.2.8–4.2.9. In $\mathrm{Spec}A$, an open subtype of an open subtype is open: by 4.2.7 the outer one is $D(f_1, \ldots, f_n)$, on each $\mathrm{Spec}A[f_j^{-1}]$ the inner one is open, hence affine-open, and gluing applies. The case $A = R$ says that open propositions are closed under dependent sums (4.2.9(a)), and applying it pointwise gives transitivity of open subtypes for every type (4.2.9(b)), schemes included, without a definition of schemes.

Lemma 4.2.11. For P closed and Q open, $(P \to Q) \Leftrightarrow \neg P \vee Q$. The paper states no axioms for it, but its proof uses Proposition 2.2.3(b), so Loc and SQC are hypotheses here.

Section 5: schemes

Small covers and local choice. A chart of $X : \mathsf{U}_l$ consists of a presentation p, an embedding $\mathrm{Sol}_p(R) \to X$, and openness of its image. A cover consists of finitely many charts and a proof that every point merely belongs to one of their images. Both $\mathrm{Cover}(X)$ and

$$\mathrm{Scheme}(X) = \|\mathrm{Cover}(X)\|$$

belong to U_l. No setness hypothesis is imposed on X: it follows from the chart embeddings and the setness of solution types. Finite choice proves equivalence with the paper's definition by affine open subtypes. The key reduction is that the large assertion that T is affine is equivalent to the small proposition $\|\sum_p T \simeq \mathrm{Sol}_p(R)\|$.

For Proposition 5.1.3, apply Z-choice on the presented algebra of each chart to a family $B : X \to \mathsf{U}_l$ with pointwise mere inhabitants. This gives finite unimodular lists of functions and sections on their principal opens. Localisation identifies each principal open with an affine type, which

can again be represented by a small presentation. Finite choice and concatenation yield merely a finite affine open cover of X with sections of B. The proof retains the standard-open refinement data, and also provides sections on the chart images. All Z-choice applications are at level l; no higher-universe choice or choice over all points of X is added.

A related lemma flattens a finite open cover by schemes to an affine open cover. It uses finite choice of the covers, concatenation, and open transitivity. It does not separately invoke Theorem 4.2.7.

Specialisation and gluing. For Lemma 5.2.1, take an affine chart containing x. If x specialises to y, the chart also contains y. Open transitivity extends opens of the chart to X, so the affine specialisation theorem gives $\neg\neg(x = y)$. The reverse implication holds for any type.

For Proposition 5.3.1, given embeddings $f : V \to X$ and $g : V \to Y$ into sets, quotient $X + Y$ by equality on each summand and by

$$\mathrm{Both}(x, y) = \sum_{v:V} (f(v) = x) \times (g(v) = y)$$

between the summands. The embedding assumptions make this an equivalence relation with proposition-valued fibres. Effectiveness of the quotient shows that its two inclusions are embeddings and identifies their intersection. Quotient descent proves that restriction

$$(P \to B) \longrightarrow \sum_{h:X\to B} \sum_{k:Y\to B} \prod_{v:V} h(f(v)) = k(g(v))$$

is an equivalence for every set $B : \mathsf{U}_l$. The universal property against arbitrary types, or sets at a separate larger level, is not claimed.

When f and g have open images, the two inclusions into P do too: this is checked on quotient representatives by proposition-valued induction. Composing the affine charts of X and Y with these open embeddings and concatenating them gives a cover of P. The quotient and its set-target universal property use no geometric axiom; openness uses the proved transitivity theorem.

Subschemes and equality types. For Propositions 5.4.2–5.4.3, restrict a scheme's affine charts to the given subtype. For a closed subtype, each restriction is affine by Lemma 4.1.4. For an open subtype of an affine type, Theorem 4.2.7 provides a finite cover by principal opens, each affine by localisation. The finite-cover flattening lemma gives schemehood in the general open case. The definition of subscheme explicitly includes proposition-valuedness of the family.

Lemma 5.5.1 uses no geometric axiom: equality in an affine type is a finite system of coordinate equations. Every closed proposition is itself affine, using a presentation with zero variables and those constant equations. For Proposition 5.5.2, choose an affine open U containing x, with witness u_x. The open proposition $U(y)$ is a scheme, and

$$\sum_{u:U(y)} (x, u_x) = (y, u)$$

is its closed subtype, where the equality is in the affine chart. It is therefore a scheme and is equivalent to $x = y$. The argument allows $U(y)$ to be empty.

Dependent sums and families (Section 5.6). Theorem 5.6.1 states that, for a scheme X and a family of schemes $Y : X \to \mathsf{U}_l$, the total space $\sum_{x:X} Y(x)$ is a scheme. The proof uses two distinct padding constructions. First suppose X is affine and each fibre has a chosen presentation. Theorem 3.3.3 merely bounds their numbers of variables by some N. Each solution set is equivalent to a closed subtype of R^N: restrict to the original variables, impose the original equations, and require all added coordinates to be zero. The numbers of equations may vary without any uniform bound. Consequently the total space is a closed subtype of the affine type $X \times R^N$, hence affine by 4.1.4. Local choice of presentation witnesses and finite-cover flattening then show that an arbitrary family of affine fibres over a scheme has a scheme total space. This last assertion does not claim affineness.

For general scheme fibres, local choice gives actual finite covers of the fibres on affine charts of the base. Theorem 3.3.3 bounds the numbers of charts locally. Pad each cover to that bound using empty affine charts, presented by the equation $1 = 0$. For each fixed chart index, the family has affine fibres, so the preceding result applies. The resulting finite open cover proves schemehood of the total space. All choices are eliminated into proposition-valued goals and all Z-choice applications remain at level l.

Univalence gives the equivalence between maps into X and X-indexed families: a map is sent to its actual fibres, and a family to its total space with its projection. Closure under dependent sums and under scheme-valued equality restricts this to the equivalence of schemes in Corollary 5.6.2. For proposition-valued families this gives 5.6.3: a subtype is a subscheme exactly when every value is a scheme. Finally, the pullback

$$\sum_{x:X} \sum_{y:Y} f(x) = g(y)$$

is a scheme by two applications of dependent-sum closure and 5.5.2, proving 5.6.4.

Scope and checks. The formal geometric proofs take Loc, SQC and Z-choice. In particular, the present proofs of 5.1.3 and 5.2.1 use open transitivity and do not recover the paper's weaker hypothesis lists (Z-choice alone and Loc/SQC, respectively). The combined entry point `schemes/Section5.seg` checks 5194 declarations with no failures in 1.30 seconds. Regressions include empty covers, empty subtypes, and the empty equality type between zero and one on the affine line, zero-constrained padding, empty pullbacks, and classification of a map with two-point fibres. All 20 expected negative tests are rejected.

Example 6.1.9: squaring on the projective line

$\mathbb{P}^1$ is the quotient of two copies of R (the charts $[a : 1]$ and $[1 : b]$) by the relation $ab = 1$. It is shown equivalent to the quotient of unimodular pairs by multiplication with units. Squaring descends to $s : \mathbb{P}^1 \to \mathbb{P}^1$, with $s([x : y]) = [x^2 : y^2]$. The two fibres are computed:

$$s^{-1}([0 : 1]) \simeq \{a : R \mid a^2 = 0\}, \qquad s^{-1}([1 : 1]) \simeq \mathsf{Bool} \quad \text{if } 2 \neq 0.$$

The first is the case $t = 0$ of a chart computation valid for every t, $s^{-1}([t : 1]) \simeq \{a \mid a^2 = t\}$, which only needs a ring. The second uses Loc, $1 \neq 0$, and that 2 is invertible. That $2 \neq 0$ implies 2 invertible uses SQC for the algebras $R/(x, y)$, given as a quotient of R, with evaluation into functions on $\{x = 0 \wedge y = 0\}$. These are the algebras used in the proof of Proposition 2.2.3.

The coordinate algebra of a fibre is treated in the setting of presentations. Let K be the algebra presented by one variable T and the equation $T^2 = a$. Then $(b, c) \mapsto b + cT$ is an equivalence $R \times R \simeq K$ compatible with the module operations, so K is free of rank two with basis $1, T$. With

SQC, the functions on $\{x : R \mid x^2 = a\}$ are exactly the $x \mapsto b + cx$. For $a = 0$ this is first-order expansion on the infinitesimal disk: every f satisfies $f(d) = f(0) + f' d$ for a unique slope f'. This uses neither Loc nor the invertibility of 2. The connection with the fibres of s computed above is not made formally.

Lemma 6.2.1: functions $\mathbb{P}^1 \to R$ are constant

The map $R \to (\mathbb{P}^1 \to R)$ sending c to the constant function is an equivalence, with inverse evaluation at $[0 : 1]$. The hypotheses are SQC for $R[X]$ (evaluation $R[X] \to (R \to R)$ is an equivalence) and SQC for $R[X, X^{-1}]$ (evaluation into functions on $\{(x, y) \mid xy = 1\}$). The two restrictions of f to the charts are merely polynomials $p(X)$ and $q(X^{-1})$. They agree on the overlap, so by injectivity for the Laurent polynomials all positive coefficients vanish and the constants agree. The coefficient computation does not use SQC. Constancy then follows by induction over the quotient $\mathbb{P}^1$.

Proposition 4.6 of [2]: line bundles on $\mathbb{P}^1$

Line is the small type of $R^\times$-torsors and $\mathbb{P}^1$ the quotient of two charts above. The map $\mathsf{Line} \times \mathbb{Z} \to (\mathbb{P}^1 \to \mathsf{Line})$, $(l_0, d) \mapsto (x \mapsto l_0 \otimes \mathcal{O}(d)(x))$, is an equivalence, from Loc, SQC and Z-choice. Its inverse sends L to $L([0 : 1])$ and the degree of L; both inverse equations are checked.

Horrocks (Lemma 2.1 of [2]). A line bundle on Spec$A$ is given by a rank-one projector E over $A[X]$. Let A be any commutative ring and f monic. If E factors after inverting f, that is $x_i y_j = f^k E_{ij}$ and $\sum_i y_i x_i = f^k$, then $E = ba$ with $\sum_i a_i b_i = 1$. No SAG axiom is used. The proof follows Appendix 1 of [2], with explicit certificates throughout:

- *Formal gcd.* A node is a pair of finite lists (I, U) of elements of A; it inverts the monoid $S(I; U) = M(U) + (I)$, which puts (I) in the Jacobson radical. Starting from f^k, the row ideal $(y_1, \ldots, y_n)$ always contains a monic polynomial. The remainders of the y_i are scanned from the top coefficient down. A coefficient a/s is tested by branching on a: either a joins I, or a joins U and becomes a unit. A unit coefficient gives, by monic lifting, a monic member of smaller degree. The recursion is on the degree bound, the position and the number of coefficients; no degree or unit is decided.
- *Leaves.* At each leaf the monic member generates the row ideal modulo (I). By the saturation calculation and a determinant-free Nakayama elimination in $A_S[X]/(h)$, it generates the row ideal itself.
- *Comaximality.* For any choice of one $s_\lambda \in S_\lambda$ per leaf, the s_λ generate the unit ideal: two children combine through $w = s^k v + (-b)^k j$, and the root monoid is $\{1\}$.
- *Gluing.* Clearing denominators at leaf i gives $d_i, \tau_i \in S_i$ and, in $A[X]$, $p_i = \sum_j y_j z_{ij}$ with $p_i = d_i h_i$ over A_{S_i} and $\tau_i y_j = p_i b_{ij}$. Over the product monoid $S_i S_j$ both monic generators generate the row ideal, so they are equal (also when their degrees differ, since then $1 = 0$); hence $u_{ij} v_{ij} (d_j p_i - d_i p_j) = 0$ with $u_{ij} \in S_i$, $v_{ij} \in S_j$. With $r_i = \prod_k u_{ik} v_{ki}$ the numerators $H_i = r_i p_i$ satisfy $s_i H_j = s_j H_i$ for $s_i = d_i r_i$, and $t_i y_j = H_i b_{ij}$ for $t_i = r_i \tau_i$. Two partitions of unity, for (s_i) and (t_i), give $g \in (y)$ with $y_j = g a_j$. The global g need not be monic.
- *Descent.* From $g = \sum_j y_j c_j$ and $y_j = g a_j$, put $b_i = \sum_j E_{ij} c_j$; cancelling the regular element f^k gives $E_{ij} = b_i a_j$ and $\sum_i a_i b_i = 1$.

Chart triviality (Corollary 2.3 of [2]). A line bundle on an affine algebra gives a rank-one projector whose fixed vectors are the global sections. A frame of L on $D(f)$, extended by Lemma 7.1.8, gives exactly the certificate above, so Horrocks makes L trivial on $\mathbb{A}^1$ when it is trivial on a monic open. Z-choice gives a polynomial neighbourhood of 0 with a frame; reversing and normalising a polynomial with invertible constant term gives a monic open in the other chart. Hence the restriction of $L : \mathbb{P}^1 \to \mathsf{Line}$ to each chart is trivial.

Laurent algebra (Lemmas 4.1 and 4.4 of [2]). On the overlap, L is then given by a function from units to units. By SQC every function on the units is merely a Laurent polynomial. Lemma 4.1 is proved for R: an invertible Laurent polynomial merely has one unit coefficient, and every other coefficient is not nonzero. Coefficients are decided only under double negation. Lemma 4.4 factors an invertible Laurent polynomial as $uX^d(1+a)(1+b)$ with $a \in XR[X]$, $b \in X^{-1}R[X^{-1}]$ nilpotent; the nilpotent part is removed one power of a single nilpotent at a time. The degree is unique, since $x^N a(x) = x^M b(1/x)$ with units a, b forces $N = M$.

Scope and checks. The Horrocks package checks in 2.79 seconds (4982 declarations), the Laurent algebra in 1.17 seconds, and the whole of Proposition 4.6 in 4.84 seconds (9773 declarations, no failures), also from an isolated copy of its sources. A false final proof is rejected.

Section 7 of [2]: line bundles on $\mathbb{P}^n$

The two definitions. $\mathbb{P}^n$ is defined as the vectors of R^{n+1} with an invertible coordinate, modulo multiplication by units; the relation is a proposition and is equivalent to proportionality. It is shown equivalent to the type of pairs of $l : \mathsf{Line}$ and a nonzero equivariant map from l to R^{n+1}. With Loc and SQC, a vector has an invertible coordinate if and only if it is nonzero. These results hold for any commutative ring.

Lemma 1.6 and Proposition 1.7. For $k \geq 2$, every function $R^k \setminus \{0\} \to R$ extends uniquely to R^k. This is proved as a Hartogs lemma for a finitely presented algebra A with regular elements $s_0, s_1, \ldots$ such that $s_1^b g = s_0^a h$ implies $s_0^a \mid g$: functions on $D(s_0) \cup D(s_1) \cup \cdots$ are elements of A. It follows from SQC for A and its localisations. Proposition 1.7, every function $\mathbb{P}^n \to R$ is constant, follows as in [2].

Lemma 7.1. For $n \geq 2$ and $p \neq q$, functions $\mathbb{P}^n \setminus \{p, q\} \to R$ are constant. A linear automorphism, built from two transpositions and Cramer's rule, moves p, q to $[e_Y], [e_Z]$. Then $\mathbb{P}^n \setminus \{[e_Y], [e_Z]\}$ lifts to the open subset $D(X) \cup D(YZ) \cup \bigcup_j D(x_j)$ of R^{n+1}, x_j the other coordinates. This family satisfies the hypotheses of the Hartogs lemma, so no gluing of chart extensions is needed. For $\mathbb{P}^n \setminus \{p\}$, a point $x \neq [e_Y]$ is apart from $[e_Z]$ or from $[e_X]$, and the two cases share the point $[e_X + e_Z]$. The statements for $\mathbb{Z}$ follow from those for R, since $\mathbb{Z}$ has decidable equality.

$\mathcal{O}(d)$ and Lemma 7.2. The fibre of $\mathcal{O}(d)$ at p is the torsor of unit-valued functions v on the representatives of p with $v(ux) = u^d v(x)$; so $\mathcal{O}(1)$ is dual to the tautological line. For $[x] \neq [y]$ the line $\mathbb{P}^1 \to \mathbb{P}^n$ sends $[a : 1]$ to $[ax + y]$ and $[1 : b]$ to $[x + by]$. The representatives of a point of $\mathbb{P}^1$ in the two charts give representatives of its image with the same transitions, so $\mathcal{O}(d)$ restricts to the $\mathcal{O}(d)$ of Proposition 4.6. With $\mathrm{Gr}(1, M)$ parametrised by $\mathbb{P}^1$ through a basis x, y of M, this is Lemma 7.2.

Theorem 7.3. For $n \geq 2$ the map $\mathsf{Line} \times \mathbb{Z} \to (\mathbb{P}^n \to \mathsf{Line})$, $(l_0, d) \mapsto (x \mapsto l_0 \otimes \mathcal{O}(d)(x))$, is an equivalence. Its inverse sends L to $L([e_Z])$ and the degree of L on the line through $[e_Y]$ and $[e_Z]$. The main step is that two line bundles L_1, L_2 on $\mathbb{P}^n$ that agree on one line are merely equal:

- For a point p, that L_1 and L_2 agree on the lines through p and x is a proposition of $x \neq p$. It is decidable, by comparing degrees on $\mathbb{P}^1$, so by Lemma 7.1 it holds for all x once it holds for one.
- Since functions $\mathbb{P}^1 \to R^\times$ are constant, an identification $L_1(p) = L_2(p)$ extends uniquely along each line through p. This identifies L_1 and L_2 on $\mathbb{P}^n \setminus \{p\}$, independently of the representatives chosen for the line.
- Doing the same at a second point q, the two identifications differ on $\mathbb{P}^n \setminus \{p, q\}$ by a function to $R^\times$, constant by Lemma 7.1. After correcting by it they glue.

This avoids $L \otimes \mathcal{O}(-d)$, so associativity and inverses for the tensor product of lines are not needed. An equality between merely equal bundles is determined by its value at one point, since functions $\mathbb{P}^n \to R^\times$ are constant.

Corollary 5.6: $\mathrm{Aut}(\mathbb{P}^n) = \mathrm{PGL}_{n+1}(R)$. For $n \geq 2$, the action of $\mathrm{PGL}_{n+1}(R)$, the linear automorphisms of R^{n+1} modulo units, on $\mathbb{P}^n$ is an equivalence onto the maps $\mathbb{P}^n \to \mathbb{P}^n$ that have an inverse, and it respects composition. The proof does not use Corollary 5.5 or tensor powers of lines.

- A map $\varphi : \mathbb{P}^n \to \mathbb{P}^m$ merely lifts to $F : R^{n+1} \setminus \{0\} \to R^{m+1} \setminus \{0\}$ with $F(ux) = u^d F(x)$: in the first definition, φ gives a line bundle with a nonzero equivariant vector; by Theorem 7.3 the bundle is merely $\mathcal{O}(d)$, and a representative x gives a frame of $\mathcal{O}(d)$ at $[x]$.
- By Lemma 1.6 a function of negative weight vanishes, so $d \geq 0$; and a function that is a unit on $R^k \setminus \{0\}$, $k \geq 2$, extends to a unit (extend it and its inverse), so it cannot have positive weight.
- For φ with inverse ψ and lifts F, G of weights a, b, $G(F(x)) = \lambda(x)x$ with $\lambda(ux)u = (u^a)^b \lambda(x)$. By the previous point $a = b = 1$, and λ is constant by Proposition 1.7.
- A function of weight 1 is additive: $f(x+ty) = f(x)+ta_1(t)$, and the chart intersection lemma of Lemma 6.2.1 makes a_1 constant, equal to $f(y)$. So F extends to a linear map, with GF and FG scalars.
- Injectivity: if $[Ax] = [Bx]$ for all x, the ratio is a function on $\mathbb{P}^n$, hence a constant unit.

Comparison with [2]. The formal proofs supply some steps that the text leaves implicit.

- Corollary 5.6 is stated without proof after Corollary 5.5. The missing steps are: the degrees d, e of an automorphism and of its inverse satisfy $de = 1$; a map of degree 1 is linear; it is invertible up to a scalar; and two matrices inducing the same map differ by a unit.
- In Corollary 5.5 the degree is not negative. This needs an argument: a function of negative weight on $R^{n+1} \setminus \{0\}$ vanishes.
- In Lemma 7.1, U_1 should be $\{x_1 \neq 0\}$. That one may assume $p \in U_0 \setminus U_1$ and $q \in U_1 \setminus U_0$ needs a change of coordinates, and $n > 1$ is where Lemma 1.6 is used. Parts (i) and (iii) need a third point, or the observation that every $x \neq p$ is apart from one of two further points.
- In Lemma 7.2 the map should be $L \mapsto l_0 \otimes L^{\otimes d}$, and for $d < 0$, $\mathrm{Hom}(L^{\otimes(-d)}, R)$. In Theorem 7.3, $n > 1$ is needed, and independence of the degree needs both that it is symmetric in p, q and that every x is apart from p or from q.

Scope and checks. The directory `pn` checks 12188 declarations with no failures in 6.6 seconds, including all of `picard`. Each of its files checks in at most 0.25 seconds. Other results of [2] are not formalised.

Section 7.1: weakly quasi-coherent modules on schemes

Modules. An R-module is the module record of the torsor files, over the same carrier and operations as R. Sections 2–5 and the torsor files use separate copies of $\mathbb{N}$ and of finite types; explicit translations between them make powers and finite sums agree. For $X : \mathsf{U}_l$ and modules M_x, $M(X) = \prod_{x:X} M_x$ with pointwise operations (7.1.1). A fraction m/g^N is a pair; $m/g^p = m'/g^{p'}$ is witnessed by K with $g^K(g^{p'}m - g^p m') = 0$. Most results below are stated with fractions; the localised module M_g is constructed at the end of this subsection, and 7.1.2 and 7.1.8 are also proved there as equivalences of modules.

A module M is *weakly quasi-coherent* (7.1.5) if, for every $g : R$, an element of M given when g is invertible is merely m/g^N, and an m that is zero when g is invertible is merely killed by a power of g. A family is weakly quasi-coherent (7.1.7) if every M_x is.

Boundedness on schemes. Proposition 3.3.4 extends to a scheme X: if $P : X \to \mathbb{N} \to \mathsf{U}_l$ is proposition-valued and closed upwards, and every $P\,x$ merely has an element, then merely one N satisfies $P\,x\,N$ for all x. Z-choice gives actual witnesses on each chart, affine boundedness bounds them there, and finite choice over the charts gives the bound. No witness is chosen at every point.

Lemma 7.1.2, for fractions. On a scheme, pointwise merely existing witnesses of $m/g^p = m'/g^{p'}$ give one exponent for all x. Over a finite open cover, local numerators with a common denominator whose fractions merely agree on overlaps merely glue to one global m, after raising the denominator by some g^d. The argument is first proved for any family of sets representing fractions, given that raising numerator and denominator preserves a fraction and that equal fractions with the same denominator have numerators that agree after finitely many steps; for modules both are proved.

Lemma 7.1.8. If X is a scheme and every M_x is weakly quasi-coherent, then for $g : X \to R$, $M(X)_g \to M(D(g))$ is bijective: a section over $D(g)$ is merely m/g^N with $m : M(X)$, and an m that vanishes on $D(g)$ is merely killed by a power of g. The first part is the gluing above; the second applies boundedness to $P\,x\,K \equiv (g(x)^K m_x = 0)$. Applied to $\sum_{x:X} U(x)$, which is a scheme by Section 5, it gives the form for an open $U \subseteq X$, which need not be affine.

Sections and images (7.1.10–7.1.12). For a constant scalar $g = r$, exchanging the arguments identifies sections on $D(g) \subseteq X$ with $(\prod_x M_x)^{D(r)}$. Thus the two clauses of 7.1.8 directly give weak quasi-coherence of $\prod_x M_x$. We prove 7.1.10 for every scheme X: the paper states it for affine X, but applies it in 7.1.12(b) to fibres that need only be schemes. Inverse image preserves weak quasi-coherence pointwise. Direct image is the product over the actual fibre, which is a scheme by 5.6.4, so the strengthened 7.1.10 proves 7.1.12(b).

Finitely presented algebras (7.1.9). First, R is weakly quasi-coherent: SQC for the presented localisation R_g represents functions on $D(g)$, and its fraction and kernel lemmas give the two clauses. For a finitely presented R-algebra A, 7.1.10 then makes $(\mathrm{Spec}A \to R)$ weakly quasi-coherent. The canonical SQC equivalence $A \simeq (\mathrm{Spec}A \to R)$ preserves zero and scalar multiplication, so the

fraction clauses transfer to the underlying R-module of A. The bundle statement follows pointwise. This proof uses the full SAG axioms through 7.1.10; it does not recover the smaller hypothesis list of the paper's SQC tensor argument. The general module tensor map in Remark 7.1.3 is not formalised.

Kernels and cokernels (7.1.6, 7.1.13). A submodule is a proposition-valued predicate closed under the operations. The quotient P/K is the effective quotient of P by $b - a \in K$, with the operations defined on representatives; the projection is linear and onto, with kernel K.

For 7.1.6, let $f : P \to Q$ be linear with P, Q weakly quasi-coherent. A section of $\ker f$ over $D(r)$ is m/r^N with $m \in P$. Then $f(m)$ vanishes on $D(r)$, so $r^K f(m) = 0$, and the section is $r^K m/r^{K+N}$ with $r^K m \in \ker f$. The zero clause is that of P.

Both remaining steps use one lemma: *if $p : P \to Q$ is linear and onto, and P and $\ker p$ are weakly quasi-coherent, then so is Q.* A section s of Q over $D(r)$ lifts to P: the preimages of $s(d)$ form a torsor under $\ker p$, and $D(r) \simeq \mathrm{Spec}R[1/r]$ is affine, so Theorem 7.3.6 gives a lift a. Then $a = m/r^N$ and $s = p(m)/r^N$. If $p(x)$ vanishes on $D(r)$, then $x \in \ker p$ there, so $x = k/r^N$ with $k \in \ker p$; $r^N x - k$ vanishes on $D(r)$, is killed by some r^K, and $r^{K+N} p(x) = 0$.

For 7.1.13, apply the lemma to $M \to f(M)$, whose kernel is $\ker f$, and then to $N \to N/f(M)$, whose kernel is $f(M)$. This avoids the paper's reduction through embeddings and $M/\ker f$, and its forward reference to the long exact sequence of 7.3.4.

The localised module. For a module M over any commutative ring and an element r, M_r is the effective quotient of pairs (m, p) by $r^K(r^q m - r^p m') = 0$ for some K. Transitivity uses $r^q(r^t m - r^p m'') = r^t(r^q m - r^p m') + r^p(r^t m' - r^q m'')$. Addition is $m/r^p + n/r^t = (r^t m + r^p n)/r^{p+t}$; it respects the relation in the first argument and is commutative on representatives, which gives the second argument. The canonical map $M_r \to M^{D(r)}$, $m/r^p \mapsto (u \mapsto u^p m)$ with u the inverse of r, is linear.

Definition 7.1.5. This map is an equivalence if and only if the two clauses used above hold at r: surjectivity is the first clause, and injectivity is the second, since m/r^p maps to 0 exactly when m vanishes on $D(r)$. So the clauses are the paper's definition.

Lemma 7.1.2. For a scheme X, modules M_x and $f : X \to R$, $M(X)$ is a module over the ring $X \to R$, and $m/f^p \mapsto (x \mapsto m(x)/f(x)^p)$ is a linear equivalence $M(X)_f \simeq \prod_{x:X} (M_x)_{f(x)}$. It is injective because pointwise exponents have a common bound (boundedness on schemes), and surjective by gluing fractions as above.

Lemma 7.1.8. If each M_x is weakly quasi-coherent, composing with the equivalences $(M_x)_{f(x)} \simeq M_x^{D(f(x))}$ gives $M(X)_f \simeq M(D(f))$. For an open $U \subseteq X$ and $f : U \to R$, apply this to the scheme $\sum_{x:X} U(x)$.

Equalities of modules. A linear bijection $A \simeq B$ gives $A = B$ in the type of modules. By the record eliminator every module is built from its fields; by induction on equivalences (univalence) the bijection may be taken to be the identity; then addition and scaling agree by linearity, zero and negation are determined by addition, and the laws are propositions. So 7.1.2 and 7.1.8 hold as equalities of modules over $X \to R$, as the paper states them.

Remark 7.1.3. Only its case $M_r \to M^{D(r)}$ is used, and that is the map above. The general map $M \otimes A \to M^{\mathrm{Spec}A}$ needs the tensor product of modules and is not formalised.

Scope and checks. The proofs take Loc, SQC and Z-choice. The file `qc-schemes/All.seg` loads Sections 2–5, the torsor files and the stated parts of Section 7.1 together: 6568 declarations, no failures, 1.93 seconds, with 22 expected negative tests rejected. Tests include positive annihilating

exponents, empty fibres and the nonzero direct image of R along the identity of the point. The file `cokernel7/All.seg` adds 7.1.6, 7.1.13 and the localised module, and loads all of Section 7.1: 7167 declarations, no failures, 3.43 seconds, with 24 expected negative tests rejected.

Lemma 7.3.5 and Theorem 7.3.6: H^1 of an affine scheme vanishes

Lemma 7.3.5. Let M be a module over a commutative ring A, and $f_1, \dots, f_n \in A$ with $\sum_k r_k f_k = 1$. An element of M_g is written x/g^p. Let $s_{ij} = m_{ij}/(f_i f_j)^N$ in $M_{f_i f_j}$ with $s_{jk} - s_{ik} + s_{ij} = 0$ in $M_{f_i f_j f_k}$. Then there are p and $V_i \in M$ with

$$V_i/f_i^p - V_j/f_j^p = s_{ij} \quad \text{in } M_{f_i f_j}.$$

The proof has three steps.

1. *One exponent.* The cocycle condition says that $f_i^N m_{jk} - f_j^N m_{ik} + f_k^N m_{ij}$ is killed by $(f_i f_j f_k)^{K_{ijk}}$. Let K bound the finitely many K_{ijk} (their sum). With $F_i = f_i^{N+K}$ and $\mu_{ij} = (f_i f_j)^K m_{ij}$,
$$F_i \mu_{jk} - F_j \mu_{ik} + F_k \mu_{ij} = 0 \quad \text{in } M.$$
2. *Powers.* The F_i are again unimodular: $\sum_k \rho_k F_k = 1$. By recursion on n, using that $a + x = 1$ gives $a(1 + x + \cdots + x^{q-1}) + x^q = 1$.
3. *Computation.* With $V_i = \sum_k \rho_k \mu_{ik}$,
$$F_j V_i - F_i V_j = \textstyle\sum_k \rho_k (F_j \mu_{ik} - F_i \mu_{jk}) = \sum_k \rho_k F_k \mu_{ij} = \mu_{ij},$$
which is the statement with $p = N + K$.

The paper writes $s_{ij} = m_{ij}/(f_i f_j)$ with one denominator for each index and uses the identity of step 1 in M itself; steps 1 and 2 are what this needs. The paper's u_i is $-V_i/f_i^p$, so that $s_{ij} = u_j - u_i$, while the proof of Theorem 7.3.6 uses $t_{ij} = u_i - u_j$; the formal statement uses the second convention. Here elements of M_g are pairs, and equality in M_g is the relation above; the localised module of Section 7.1 is not needed.

Theorem 7.3.6. R is a commutative ring, X a type and M_x an R-module for $x : X$. Functions $X \to R$ play the role of A (this is SQC), and $D(g)$ is the set of x where $g(x)$ is invertible. The hypotheses are:

- *Loc*: $1 \neq 0$, and if $a + b$ is invertible, then merely a or b is;
- *Z-choice*: if every B_x is merely inhabited, then merely there are unimodular $f_1, \dots, f_n : X \to R$ and sections of B over each $D(f_k)$;
- *weak quasi-coherence*, in the form used, Lemma 7.1.8 for $U = X$: for $g : X \to R$ the map $M(X)_g \to M(D(g))$ is bijective. On elements: a section over $D(g)$ is merely m/g^N for a global m, and a global m that vanishes on $D(g)$ is merely killed by a power of g.

Statement: every M-torsor T on X merely has a section, that is, merely equals the trivial torsor: $H^1(X, M) = 0$.

Proof.

- Z-choice gives sections s_i of T over $D(f_i)$, and the differences $t_{ij} = s_i - s_j$ are defined on $D(f_i f_j)$.
- By weak quasi-coherence and finite choice, merely $t_{ij} = m_{ij}/(f_i f_j)^{N_{ij}}$; a common bound N gives one exponent.

- On $D(f_if_jf_k)$ the numerator $f_i^N m_{jk} - f_j^N m_{ik} + f_k^N m_{ij}$ is $(f_if_jf_k)^N(t_{jk} - t_{ik} + t_{ij}) = 0$. By weak quasi-coherence and finite choice, it is merely killed by powers of $f_if_jf_k$ on all of X.
- The hypotheses of Lemma 7.3.5 are now identities at each point of X. At each x the lemma, for R and M_x, gives p and V_i; with $u_i = V_i/f_i^p$ on $D(f_i)$, $u_i - u_j = t_{ij}$ on $D(f_if_j)$.
- The corrected sections $(-u_i) \cdot s_i$ agree on overlaps. They glue (Lemma 1.2.2): at x, the elements equal to all of them form a proposition, inhabited because the $D(f_i)$ cover X (unimodularity and Loc for finite sums).

In the paper's proof, $M(U_i \cap U_j)_{f_if_j}$ should read $M(X)_{f_if_j}$. The two files have 870 lines.

The paper's hypotheses. For X affine, Section 7.1 removes the third hypothesis. The functions $x \mapsto x(f_i)$ on $X \simeq \mathrm{Spec}A$ turn the Z-choice axiom for A into the form above, and Loc gives the first item. Lemma 7.1.8 gives the third from weak quasi-coherence of each M_x. Hence: if X is affine, each M_x is weakly quasi-coherent, and Loc, SQC and Z-choice hold, then every M-torsor on X merely has a section. This is the paper's Theorem 7.3.6.

Lemma 7.4.2: cocycles and torsors with local sections

Let A be an abelian group and L a type with $\|L\|$. An A-torsor is a set T with a map $A \times T \to T$ such that $(a+b)\cdot t = a \cdot (b \cdot t)$ and $0 \cdot t = t$; for all t, t' there is exactly one a with $a \cdot t = t'$, written $t' - t$; and $\|T\|$. An L-cocycle is $c : L \times L \to A$ with $c_{ij} + c_{jk} = c_{ik}$. The formal statement is an equivalence

$$\textstyle\sum_{T:A\text{-torsor}}(L \to T) \;\simeq\; L\text{-cocycles}.$$

No axiom of [1] is used. Here is the formal proof, written in the usual style.

From torsors to cocycles. Given (T, u) with $u : L \to T$, let $c_{ij} = u_i - u_j$. This is a cocycle, since $(c_{ij} + c_{jk}) \cdot u_k = c_{ij} \cdot (c_{jk} \cdot u_k) = c_{ij} \cdot u_j = u_i$, and $u_i - u_k$ is the only element with this property.

From cocycles to torsors. Given c, let

$$T_c = \{v : L \to A \mid v_m = v_k + c_{km} \text{ for all } k, m\}, \qquad (a \cdot v)_k = a + v_k, \qquad \hat{u}_i = (k \mapsto c_{ik}).$$

We have $\hat{u}_i \in T_c$ since $c_{im} = c_{ik} + c_{km}$. T_c is a torsor: it is a set, as a subset of A^L; the action laws hold pointwise. For $v, w \in T_c$, the existence and uniqueness of $w - v$ is a proposition, so we may assume $k_0 : L$. Then $a = w_{k_0} - v_{k_0}$ works: $a + v_m = (w_{k_0} - v_{k_0}) + (v_{k_0} + c_{k_0m}) = w_{k_0} + c_{k_0m} = w_m$. It is unique, since $a' + v_{k_0} = w_{k_0}$ forces $a' = a$. Finally, $\|L\|$ and $i \mapsto \hat{u}_i$ give $\|T_c\|$.

One composite. The cocycle of $(T_c, \hat{u})$ is c: we need $c_{ij} \cdot \hat{u}_j = \hat{u}_i$, that is $c_{ij} + c_{jk} = c_{ik}$ for all k.

The other composite. Let c be the cocycle of (T, u) and

$$e : T \to T_c, \qquad e(t) = (k \mapsto t - u_k).$$

Then $e(t) \in T_c$, since $t - u_m = (t - u_k) + (u_k - u_m)$; e is equivariant, since $(a \cdot t) - u_k = a + (t - u_k)$; and $e(u_i) = \hat{u}_i$ by definition. e is a bijection. It is a proposition that each fibre has exactly one element, so we may assume $k_0 : L$. The element $y_{k_0} \cdot u_{k_0}$ is mapped to y, and $e(t) = y$ gives $t = (t - u_{k_0}) \cdot u_{k_0} = y_{k_0} \cdot u_{k_0}$. By univalence, an equivariant bijection preserving the points is an

equality $(T, u) = (T_c, \hat{u})$: by induction on equivalences it is enough to take e the identity. Then the two actions are equal by function extensionality, the torsor laws are propositions, and the points agree.

Comparison with the paper.

- The paper takes $T_c = \{u \mid u_i - u_j = c_{ij}\}$ with points $k \mapsto c_{ki}$. Its own computation gives the cocycle $\tilde{c}_{ij} = c_{ji}$, so (T_c, c) is a preimage of the transposed cocycle $-c$, not of c. The definitions above correct this.
- The paper does not check that T_c is a set with a free and transitive action. This is the second place where a point of L is assumed.
- The paper shows that the left side is a set, that the map is an embedding (comparing two arbitrary (T, u) and (T', u')), and that it is surjective. Its map $T \to T'$ depends on a chosen $i : L$. The formal proof gives an inverse instead, and compares each (T, u) only with $(T_c, \hat{u})$. The map e needs no chosen point, and the left side being a set is not needed.
- "$(T, u) = (T', u')$ is equivalent to an isomorphism" is one line in the paper. In the formal proof it is the only use of univalence, and only one direction is needed.
- The axiom $0 \cdot t = t$ is part of the definition, but no step of the proof uses it. The left side lies in a larger universe than the cocycles.

Appendix A shows how these steps read in the formal text.

Theorem 7.4.7: 2-cocycles and gerbes with local objects

Definitions. An A-gerbe is a type G that is a 1-type, merely inhabited and connected ($\|x = y\|$ for all x, y), with for each x a group isomorphism $\beta_x : (x = x) \to A$, the band. Being a 1-type follows from the band in [1]; it is included as a field, as in the note *About torsors and gerbes and liens*, and is a proposition, so the type of gerbes is the same. The compatibility of the bands with conjugation, $\beta_y(q^{-1}\ell q) = \beta_x(\ell)$ for $q : x = y$, is not part of the definition: it holds for any function of x, by path induction on q. An L-2-cocycle is $c : L^3 \to A$ with $c_{ijk} + c_{ikm} = c_{jkm} + c_{ijm}$, which is the condition of [1] rearranged. The statement is that

$$\textstyle\sum_{G:A\text{-gerbe}} \sum_{u:L\to G} \prod_{i,j}(u_i = u_j) \;\longrightarrow\; L\text{-2-cocycles}, \qquad c_{ijk} = \beta_{u_i}(p_{ij}\, p_{jk}\, p_{ik}^{-1}),$$

is an equivalence, for $\|L\|$. The gerbes are in the same universe as L and A. The paper gives no proof, only a reference to Deligne's construction [3, 5.3].

Torsors. Paths between two torsors are the same as equivariant maps: the type of pairs (Y, f) with $f : X \to Y$ equivariant is contractible, since an equivariant map of torsors is a bijection and by induction on equivalences. Equivariant maps $X \to X$ are the translations, so $\mathrm{Aut}(X) \simeq A$. Hence the type of torsors is a gerbe, the trivial gerbe. It lies one universe higher, but its paths are small. The image K of the single torsor A, given by replacement, is small, and contains every torsor since all torsors are merely isomorphic. K is the small trivial gerbe.

Paths in a gerbe. In a gerbe the paths $x = y$ form an A-torsor, with $a \cdot h = \beta_x^{-1}(a)\, h$. A map of gerbes which preserves the bands is therefore an equivalence: on paths it is an equivariant map of torsors, hence a bijection, so the map is an embedding; and it is surjective because the target is connected.

The cocycle. c is a 2-cocycle: both sides of the cocycle law are $\beta_{u_i}(p_{ij}p_{jk}p_{km}p_{im}^{-1})$, after moving c_{jkm} from u_j to u_i by conjugation with p_{ij}.

The proof. Being an equivalence is a proposition, so we may assume a point $i_0 : L$. We show that each fibre is contractible.

- *Centre.* Take K, with all points equal to the base point and as path from i to j the loop of band c_{i_0ij}. Its cocycle is $c_{i_0ij} + c_{i_0jk} - c_{i_0ik}$, which is c_{ijk} by the cocycle law at (i_0, i, j, k).
- *Contraction.* Given (G, u, p) with cocycle c, map $y : G$ to the torsor of paths $u_{i_0} = y$, viewed in K. This map preserves the bands: a loop ℓ at y acts on paths by composition, which is the translation by $\beta_y(\ell)$. So it is an equivalence. It sends u_i to the base point, using $h \mapsto \beta(h\, p_{i_0i}^{-1})$, and p_{ij} to the loop of band c_{i_0ij}. A structure identity principle, by induction on equivalences, turns this into an equality with the centre.

Comparison with Deligne. Deligne's inverse is the gerbe of families of torsors P_i with isomorphisms p_{ij} satisfying the cocycle condition twisted by c. The formal proof does not need it: with a point of L, which may be assumed, every gerbe with points is identified with the trivial one, and the comparison map is Deligne's map $P \mapsto \mathrm{Hom}(Q_i, P)$ at the single index i_0. The proof is about 1400 lines; most of it is the path algebra of the trivialisation.

Theorem 7.4.8: Čech H^2 of a 2-acyclic cover

Changing the chosen paths. Let (G, u, p) be a gerbe with points and paths, with cocycle c, and b a 1-cochain. Replacing each p_{ij} by $b_{ij} \cdot p_{ij}$ (composition with the loop of band b_{ij}) keeps the gerbe and the points, and the new cocycle is

$$c_{ijk} + (\delta b)_{ijk}, \qquad (\delta b)_{ijk} = b_{ij} + b_{jk} - b_{ik}.$$

The proof takes place in the torsors of paths: c_{ijk} is the difference from p_{ik} to $p_{ij}p_{jk}$, and prefixing a path is equivariant ($q\,(a \cdot h) = a \cdot (q\,h)$, by the compatibility of the band with conjugation). Hence $(b_{ij}p_{ij})(b_{jk}p_{jk}) = (b_{ij} + b_{jk} + c_{ijk}) \cdot p_{ik}$, and differences in a torsor are unique.

Objects with paths are coboundaries. For $\|L\|$,

$$\textstyle\sum_{y:G} \prod_i (u_i = y) \;\simeq\; \sum_b (c + \delta b = 0),$$

by $(y, r) \mapsto b$, b_{ij} the difference from p_{ij} to $r_i\, r_j^{-1}$. The paths r_i are part of the data on the left, so nothing is chosen; a point of L is assumed only to prove that the map is an equivalence, a proposition. So c is a coboundary exactly when merely there is an object with paths to all the u_i.

Setting. X is a type with a finite cover $U_1, \ldots, U_n$ (every x is merely in some U_i) and F a family of abelian groups over X. With $L_x = \sum_i U_i(x)$, a Čech 2-cocycle is an L_x-2-cocycle for each x, as in the paper's proof of Theorem 7.4.5. For a cocycle c, G_c is the family of gerbes given at each x by the inverse of Theorem 7.4.7. Two cocycles are cohomologous when they differ by δb; this is an equivalence relation.

Acyclicity. The cover is 2-acyclic when $H^1(U_i \cap U_j, F) = 0$ for all i, j (including $i = j$) and $H^2(U_i, F) = 0$ for all i, that is, every family of torsors over $U_i \cap U_j$ is merely equal to the trivial family, and every family of gerbes over U_i merely equal to the trivial family. This is equivalent to: such families merely have a section. Pointwise, a torsor has an element if and only if it is equal to A, and a gerbe has an object if and only if it is equal to the trivial gerbe (the trivialisation of Theorem 7.4.7 with one point); for families both directions are pointwise, by function extensionality.

Finite choice. Acyclicity gives, for each i, merely a section over the whole U_i, and for each i, j merely a section over the whole $U_i \cap U_j$. Since the cover is finite, these finitely many mere sections are combined into one by finite choice,

$$\textstyle\prod_{i:\mathrm{Fin}\, n} \|P_i\| \to \|\prod_{i:\mathrm{Fin}\, n} P_i\|,$$

proved by recursion on n (at higher type). No choice over the points of X is needed: the sections are given on whole opens by acyclicity.

The theorem. Let $\check{H}^2$ be the set of cocycles modulo being merely cohomologous, and

$$H^2(X, F) = \|\textstyle\prod_x F_x\text{-gerbe}\|_0,$$

the families of gerbes modulo mere equality; both are effective quotients constructed from replacement. The map $c \mapsto [G_c]$ induces $\check{H}^2 \simeq H^2(X, F)$, an equivalence of sets:

- *well defined*: cohomologous cocycles give the same family, by the change of paths and Theorem 7.4.7;
- *injective*: if G_c and $G_{c'}$ are merely equal, carry the points and paths of c' to the gerbes of c; acyclicity on the U_i gives paths between the two systems of points, conjugating by them does not change the cocycle, and the change of paths gives $c' = c + \delta b$;
- *surjective*: for a family of gerbes, acyclicity gives local objects on the U_i and paths on the $U_i \cap U_j$, and Theorem 7.4.7 gives the cocycle.

No separate gluing lemma for 1-types is needed: Theorem 7.4.7 already constructs the gerbe of a cocycle at each point.

Addition. The paper states an equality of groups. Addition on H^2 is defined from the sum of gerbes, constructed in three steps.

- *Sum of torsors.* For A-torsors T, S, the pairs (Z, φ) with $\varphi : T \times S \to Z$ satisfying $\varphi(a \cdot t, s) = a \cdot \varphi(t, s) = \varphi(t, a \cdot s)$ form a contractible type (assume points t_0, s_0; the centre is A with $(t - t_0) + (s - s_0)$). Its centre is $T \otimes S$ with $q : T \times S \to T \otimes S$, $q(a \cdot t, b \cdot s) = (a + b) \cdot q(t, s)$, and $\mathrm{Hom}(T \otimes S, Z)$ is equivalent to the bilinear maps.
- *Sum of gerbes.* For gerbes G_1, G_2 and $H = G_1 \times G_2$, let $F_g(h) = (h_1 = g_1) \otimes (h_2 = g_2)$, a family of torsors on H (loops at the source act, so moving the target along a loop ℓ is the translation by $+\beta(\ell)$). $G_1 + G_2$ is the image of $g \mapsto F_g$, small by replacement, with $\mu : G_1 \times G_2 \to G_1 + G_2$. A loop of F_g is a function $H \to A$; since H is connected and inhabited, $c \mapsto (\lambda h.c)$ is an equivalence $A \simeq (H \to A)$, and the band is the constant value, with no point chosen. Then $\beta(\mu(\ell_1, \ell_2)) = \beta(\ell_1) + \beta(\ell_2)$.
- *Cocycles.* For systems (u, p) and (u', p') with cocycles c, d, the points $\mu(u_i, u'_i)$ and paths $\mu(p_{ij}, p'_{ij})$ have cocycle $c + d$. Hence the family of gerbes of $c + d$ is equal to the sum of those of c and d.

The zero cocycle is realised by the trivial gerbe with constant points and identity paths, so the zero of H^2 is the class of the trivial family. The group laws of $\check{H}^2$ hold pointwise on cocycles, and are transferred to H^2 through the equivalence, which preserves zero and addition. So $\check{H}^2 \simeq H^2(X, F)$ as abelian groups.

Refinement. If a finite cover $V_1, \ldots, V_{n'}$ refines $U_1, \ldots, U_n$ through ρ with $V_j \subseteq U_{\rho(j)}$, cocycles pull back along $(j, v) \mapsto (\rho(j), v)$. Restricting the points and paths of a gerbe pulls back its cocycle, so the family of gerbes of the pulled-back cocycle is the same family. Hence, for 2-acyclic covers, the refinement map on Čech classes commutes with the two comparison maps to $H^2(X, F)$; it is therefore an additive isomorphism, independent of ρ, and refinement satisfies the identity and composition laws (by naturality and injectivity of the comparison maps). Any two 2-acyclic covers are compared canonically through $H^2(X, F)$. A common refinement receives the two Čech refinement maps whether or not it is acyclic (the pull-back of cocycles and of coboundaries uses no acyclicity); it is 2-acyclicity of the refinement that makes its own comparison map an isomorphism, and so lets the comparison of the two covers be read through it. In the formal text the quotient $\check{H}^2$ is built inside the segment that assumes acyclicity, so the refinement map on classes is stated there for 2-acyclic covers.

4 Summary

directory	content	hypotheses	lines	time
`replacement`	truncation, quotients	—	165	0.02 s
`sag`	Lemma 2.2.4	$1 \neq 0$, Nullst. for $R[X]/(f)$	807	0.03 s
`torsor`	7.3.5, 7.3.6, 7.4.2, 7.4.7, 7.4.8	7.3.6: Loc, Z-choice, WQC	4922	0.97 s
`bounded`	Theorem 3.3.3	$1 \neq 0$, two halves of SQC	428	0.07 s
`projective`	Ex. 6.1.9, Lemma 6.2.1	Loc, SQC instances	1141	0.11 s
`polynomial`	$A[X]$, universal property	—	923	0.11 s
`presentations`	presentations, Prop. 2.2.2, Thm. 3.3.3, Ex. 6.1.9 (algebra)	Loc, uniform SQC	1473	0.40 s
`sag-uniform`	Lemma 2.2.4	Loc, uniform SQC	439	0.32 s
`tensor`	3.2.1, 3.2.2, 3.3.1–3.3.3	3.2.2, 3.3: uniform SQC; 3.3.3: $1 \neq 0$	825	0.33 s
`affine`	3.1, 3.3.4; axioms of 2.1	Loc, SQC; 3.3.4: Z-choice	1835	0.65 s
`opens`	2.2.3(b),(c), 4.2	Loc, SQC; 4.2.7–4.2.9: Z-choice	686	0.89 s
`closed`	1.3.8, 4.1	4.1.2–3: SQC; 4.1.4: all three	1667	0.99 s
`schemes`, `gluing`	5 (scope above)	Loc, SQC, Z-choice	2263	1.30 s
`bridge7`, `qc-schemes`	7.1.1, 7.1.2 (fractions), 7.1.5, 7.1.7–7.1.12; 7.3.6	Loc, SQC, Z-choice	1033	1.93 s
`cokernel7`	7.1.2, 7.1.5, 7.1.6, 7.1.8, 7.1.13	Loc, SQC, Z-choice	1266	3.43 s
`horrocks`	[2] Lemma 2.1 (rank one)	—	5276	3.17 s
`laurent`	[2] Lemmas 4.1, 4.4	Loc, SQC	1217	1.17 s
`picard`	[2] Cor. 2.3, Prop. 4.6	Loc, SQC, Z-choice	2963	5.64 s
`pn`	[2] Sec. 1, 1.6, 1.7, 7.1–7.3, Cor. 5.6	Loc, SQC; 7.2, 7.3, 5.6: Z-choice	3017	6.61 s

Lines count the files of each directory, not the files it shares with others by links; times are for checking everything a directory imports. `sag-uniform` combines the proof of `sag` with the Nullstellensatz of `presentations`.

Loc, SQC and Z-choice are hypotheses, not theorems of the foundational axioms. For affine X, Theorem 7.3.6 needs only these three and weak quasi-coherence of each M_x, as in the paper. Their

interpretation in a model is not formalised here. Later results beyond those listed above are not formalised.

A Excerpts of the formal proof of Lemma 7.4.2

The file is `primrec/torsor/Torsor.seg`, about 340 lines. The excerpts below are copied from it. Only `Laws` and `Coc` are shortened (the file names the intermediate Σ-types), and `...` marks an omitted proof. In `Unit`, `(x : A) * B` is a Σ-type, `w.1`, `w.2` are its projections, `Id` is the identity type, and `calc A { a =< p > b =< q > c }` is an equational chain. `Tr.Trunc[X].carrier` is the truncation $\|X\|$ obtained from replacement, and `univ` is the univalence parameter.

Torsors, cocycles and differences. A torsor structure is the action with its laws. Free and transitive is the contractibility of $\sum_a a \cdot t = t'$. The difference $t' - t$ (`diff`) is the centre of contraction, and `uniq` says that it is the only solution.

```
AD (T : U l) (act : A -> T -> T) : U l =
  (a b : A) (t : T) -> Id T (act (add a b) t) (act a (act b t)) ;
ZE (T : U l) (act : A -> T -> T) : U l = (t : T) -> Id T (act o t) t ;
FT (T : U l) (act : A -> T -> T) : U l =
  (t t' : T) -> isContr ((a : A) * Id T (act a t) t') ;
Laws (T : U l) (act : A -> T -> T) : U l = (sT : isSet T) * Rest T act ;
                                    -- Rest: AD, ZE, FT and ||T||
TorStr (T : U l) : U l = (act : A -> T -> T) * Laws T act ;
Left : U l+ = (T : U l) * (s : TorStr T) * (L -> T) ;
Coc : U l = (c : L -> L -> A) * ((i j k : L) -> Id A (add (c i j) (c j k)) (c i k)) ;

diff (t t' : T) : A = (ft t t').1.1 ;
dact (t t' : T) : Id T (act (diff t t') t) t' = (ft t t').1.2 ;
uniq (t t' : T) (a : A) (p : Id T (act a t) t') : Id A (diff t t') a =
  ap ((b : A) * Id T (act b t) t') A (\ w -> w.1) (ft t t').1 (a, p) ((ft t t').2 (a, p)) ;
```

From torsors to cocycles. The cocycle law is `uniq` applied to the chain $(c_{ij} + c_{jk}) \cdot u_k = c_{ij} \cdot (c_{jk} \cdot u_k) = c_{ij} \cdot u_j = u_i$ (here `w = (T, s, u)`).

```
c (i j : L) : A = diff (w.2.2 j) (w.2.2 i) ;
coc (i j k : L) : Id A (add (c i j) (c j k)) (c i k) =
  symH (uniq (w.2.2 k) (w.2.2 i) (add (c i j) (c j k)) (calc w.1 {
    act (add (c i j) (c j k)) (w.2.2 k)
      =< ad (c i j) (c j k) (w.2.2 k) > act (c i j) (act (c j k) (w.2.2 k))
      =< ap w.1 w.1 (act (c i j)) (act (c j k) (w.2.2 k)) (w.2.2 j) (dact (w.2.2 k) (w.2.2 j))
      > act (c i j) (w.2.2 j)
      =< dact (w.2.2 j) (w.2.2 i) > w.2.2 i })) ;
```

From cocycles to torsors. T_c, its action and its points:

```
Cond (v : L -> A) : U l = (k m : L) -> Id A (v m) (add (v k) (c.1 k m)) ;
Tc : U l = (v : L -> A) * Cond v ;
actc (a : A) (v : Tc) : Tc =
  (\ k -> add a (v.1 k),
   \ k m -> transH (ap A A (add a) (v.1 m) (add (v.1 k) (c.1 k m)) (v.2 k m))
                   (symH (assoc a (v.1 k) (c.1 k m)))) ;
uc (i : L) : Tc = (\ k -> c.1 i k, \ k m -> symH (c.2 i k m)) ;
```

Free and transitive is proved by `ftk k0 v w`, given a point `k0`, and the point is obtained from $\|L\|$ (`hL`) because the goal is a proposition:

```
ft-c : FT Tc actc = \ v w ->
  Tr.Trunc[L].induction @l (\ x -> isContr ((a : A) * Id Tc (actc a v) w))
    (\ x -> isProp-isContr ((a : A) * Id Tc (actc a v) w)) (\ k0 -> ftk k0 v w) hL ;
```

The map e. $e(t) = (k \mapsto t - u_k)$, with the proof that it lies in T_c. The proof that e is a bijection (`fib`) uses a point of L in the same way as `ft-c`.

```
e (t : T) : PsiM[phi w].Tc =
  (\ k -> diff (u k) t,
   \ k m -> uniq (u m) t (add (diff (u k) t) (c k m)) (calc T {
     act (add (diff (u k) t) (c k m)) (u m)
       =< ad (diff (u k) t) (c k m) (u m) > act (diff (u k) t) (act (c k m) (u m))
       =< ap T T (act (diff (u k) t)) (act (c k m) (u m)) (u k) (dact (u m) (u k))
       > act (diff (u k) t) (u k)
       =< dact (u k) t > t })) ;
eqv : Equiv T PsiM[phi w].Tc =
  (e, qinv T PsiM[phi w].Tc e (\ y -> (fib y).1.1) (\ y -> (fib y).1.2) ...) ;
path : Id Left w (psi (phi w)) =
  sip w.1 w.2.1 w.2.2 PsiM[phi w].Tc eqv PsiM[phi w].str PsiM[phi w].uc (e-act, e-pt) ;
```

Univalence. Induction on equivalences, from univalence, and its use: an equivariant equivalence preserving the points gives an identification in `Left`. It is enough to prove this for the identity (`sip-base`), by function extensionality on the action and propositionality of the laws.

```
eqind (T : U l) (P : (T' : U l) -> Equiv T T' -> U l+) (d : P T (idEquiv T))
  (T' : U l) (e : Equiv T T') : P T' e =
  Prelude[l+].subst (Equiv T T') (P T')
    (Axioms[l].idtoeqv T T' ((univ @l T T').sec e)) e ((univ @l T T').sec-eq e)
    (idJ (U l) T (\ X q -> P X (Axioms[l].idtoeqv T X q)) d T' ((univ @l T T').sec e)) ;

SipMot (T : U l) (s : TorStr T) (u : L -> T) (T' : U l) (e : Equiv T T') : U l+ =
  (s' : TorStr T') (u' : L -> T') -> Eqv T T' s s' u u' e
  -> Id Left (T, s, u) (T', s', u') ;
sip (T : U l) (s : TorStr T) (u : L -> T) (T' : U l) (e : Equiv T T')
  : SipMot T s u T' e = eqind T (SipMot T s u) (sip-base T s u) T' e ;
```

The lemma.

```
lemma-7-4-2 : Prelude[l+].Equiv Left Coc =
  (phi, Prelude[l+].qinv Left Coc phi psi phipsi psiphi) ;
```

References

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